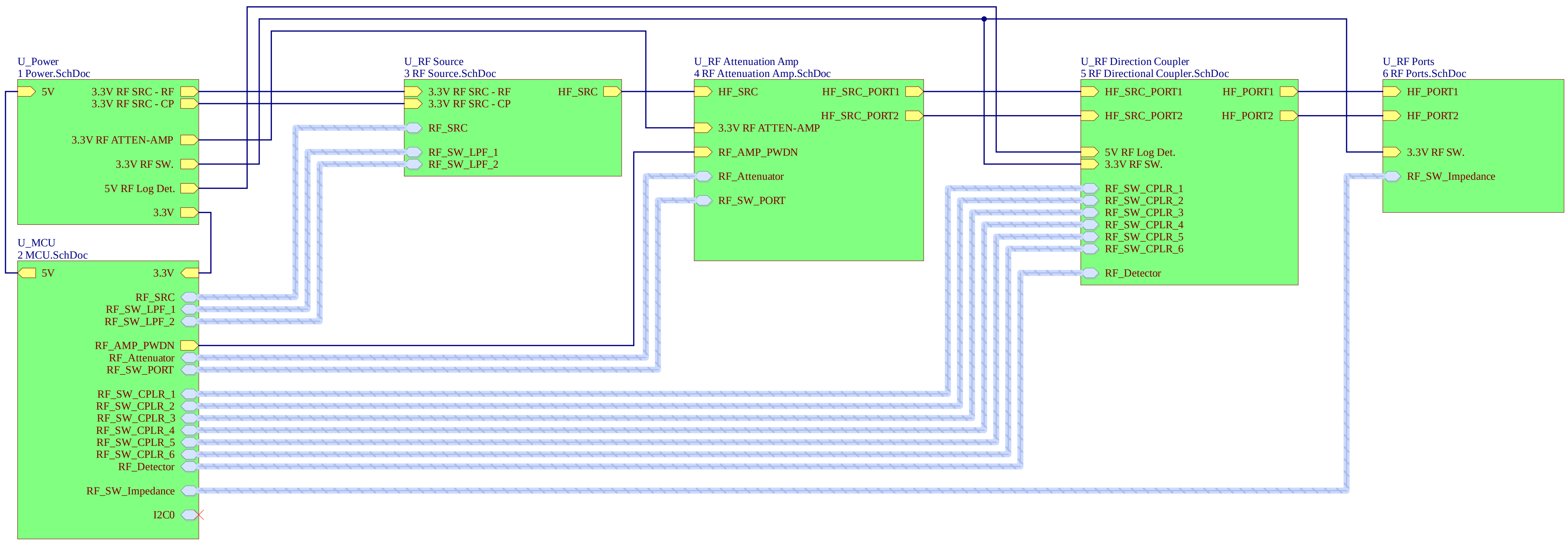

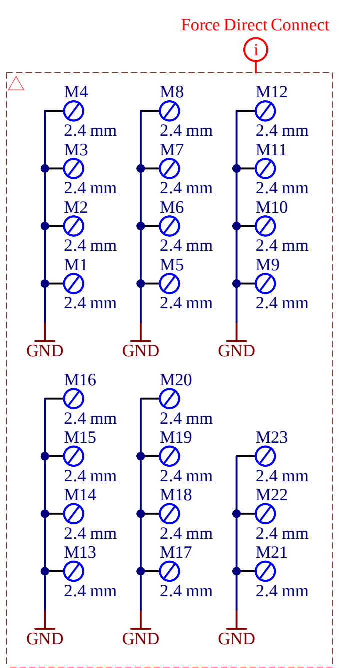

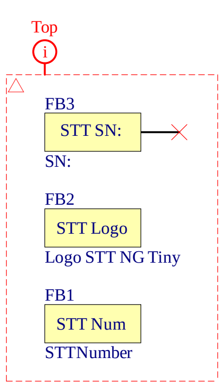

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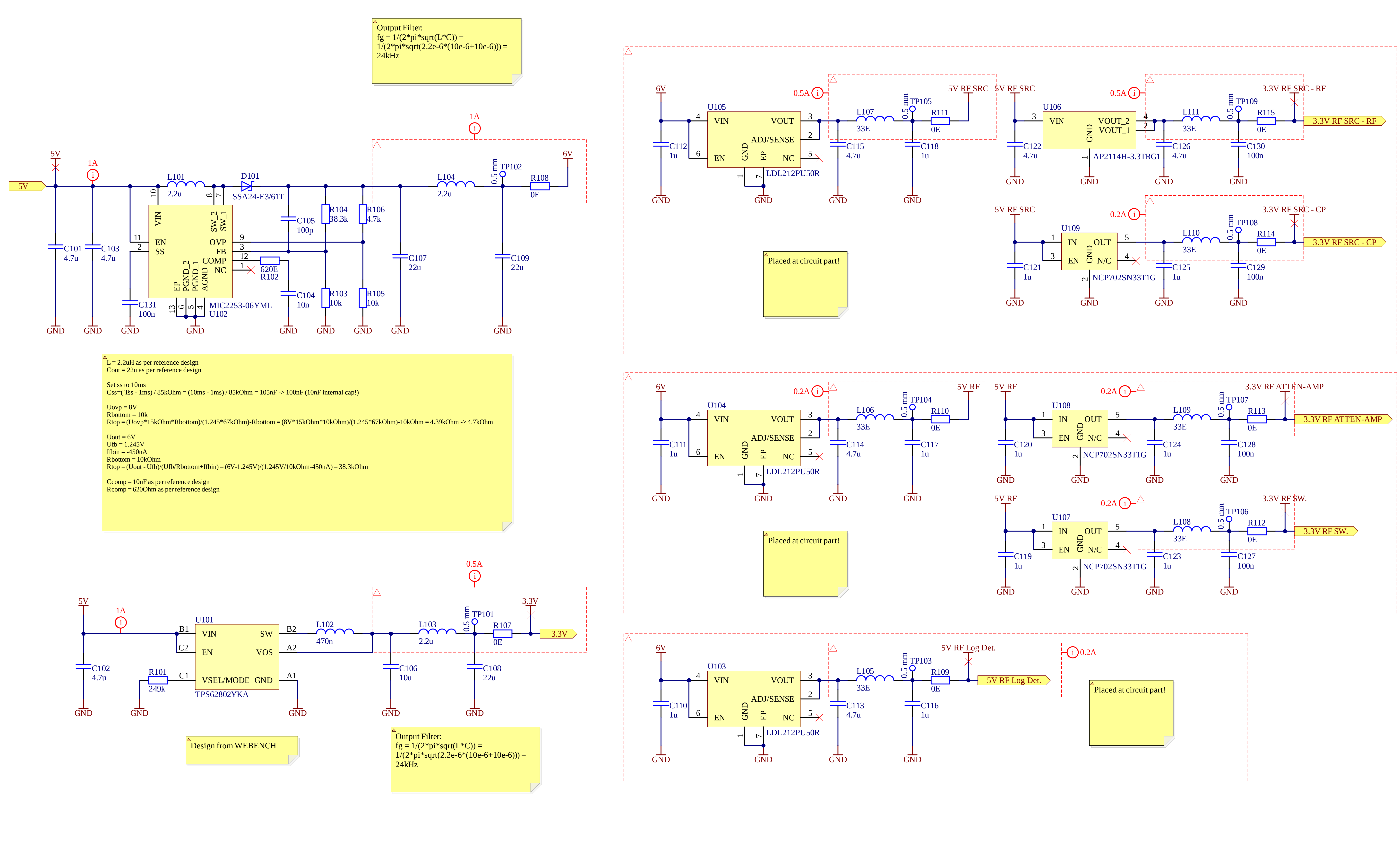
Gibb, Bern

Project:	SNA	Number:	SNA	Git:	1524f681
Board:	SNA	Variant:	EMS	Version:	1.0
Page:	Overview	Release:	03.09.2026	Sheet 1 of 7	Design: ML
				Size: A3	Check: RM

File: https://github.com/ml3rc/SNA.git 0 Overview.SchDoc



Output Filter:
fg = 1/(2*pi*sqrt(L*C)) =
1/(2*pi*sqrt(2.2e-6*(10e-6+10e-6))) =
24kHz

L = 2.2uH as per reference design
Cout = 22u as per reference design

Set ss to 10ms
Css=(Tss - 1ms) / 85kOhm = (10ms - 1ms) / 85kOhm = 105nF -> 100nF (10nF internal cap!)

Uovp = 8V
Rbottom = 10k
Rtop = (Uovp*15kOhm*Rbottom)/(1.245*67kOhm)-Rbottom = (8V*15kOhm*10kOhm)/(1.245*67kOhm)-10kOhm = 4.39kOhm -> 4.7kOhm

Uout = 6V
Ufb = 1.245V
Ifbin = -450nA
Rbottom = 10kOhm
Rtop = (Uout - Ufb)/(Ufb/Rbottom+Ifbin) = (6V-1.245V)/(1.245V/10kOhm-450nA) = 38.3kOhm

Ccomp = 10nF as per reference design
Rcomp = 620Ohm as per reference design

Design from WEBENCH

Output Filter:
fg = 1/(2*pi*sqrt(L*C)) =
1/(2*pi*sqrt(2.2e-6*(10e-6+10e-6))) =
24kHz

For current specs see power blockdiagramm
20mil -> 1A (2.14A)
15mil -> 0.5A (1.79A)
12mil -> 0.2A (1.56A)

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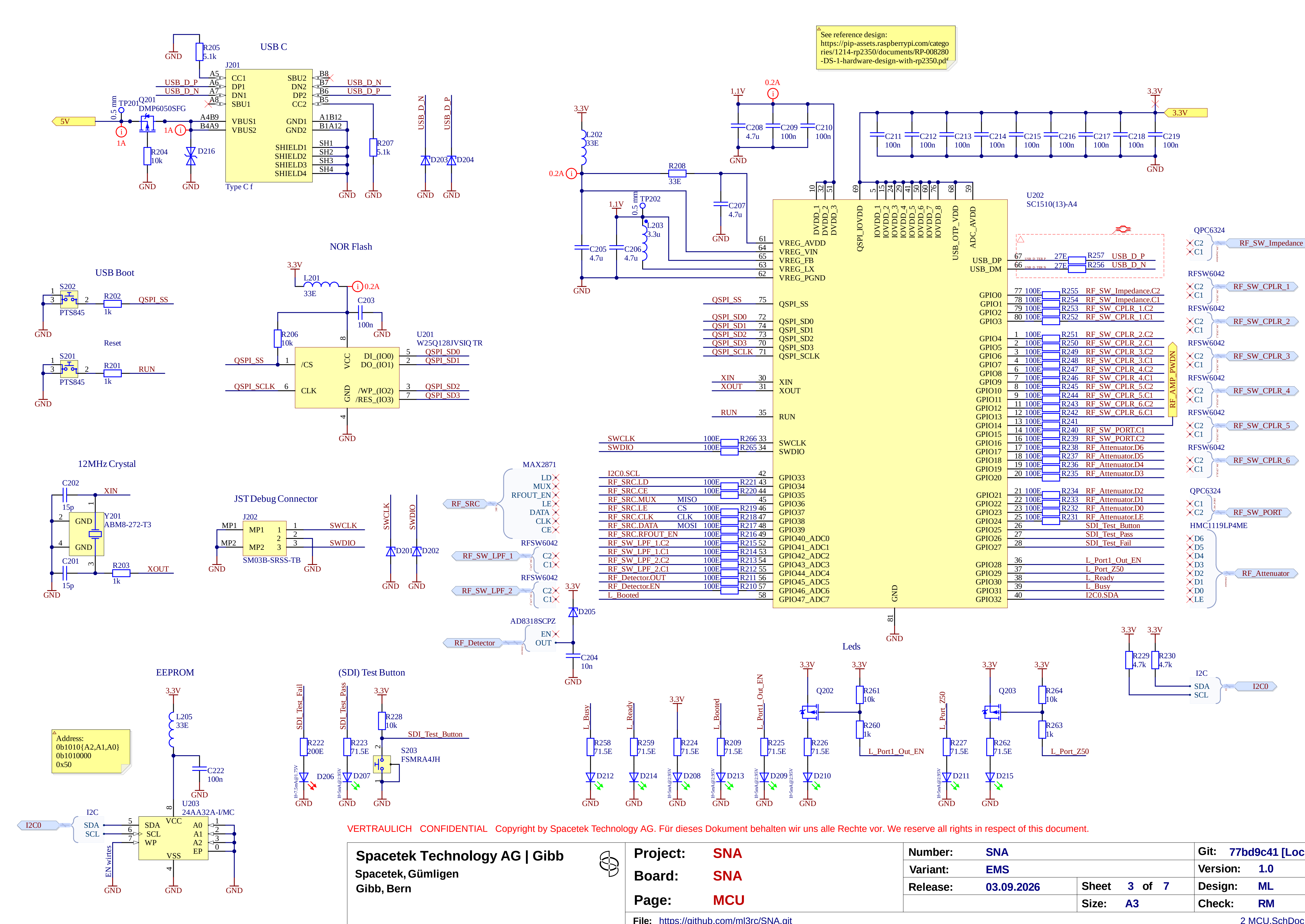
Project:	SNA	Number:	SNA
Board:	SNA	Variant:	EMS
Page:	Power	Release:	03.09.2026
Sheet	2 of 7	Size:	A3
Git:	77bd9c41	Version:	1.0
Design:	ML	Check:	RM

File: https://github.com/ml3rc/SNA.git 1 Power.SchDoc



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Gibb, Bern

Project:	SNA	Number:	SNA
Board:	SNA	Variant:	EMS
Page:	MCU	Release:	03.09.2026

Sheet 3 of 7

Size: A3

Git: 77bd9c41 [Locally M

Version: 1.0

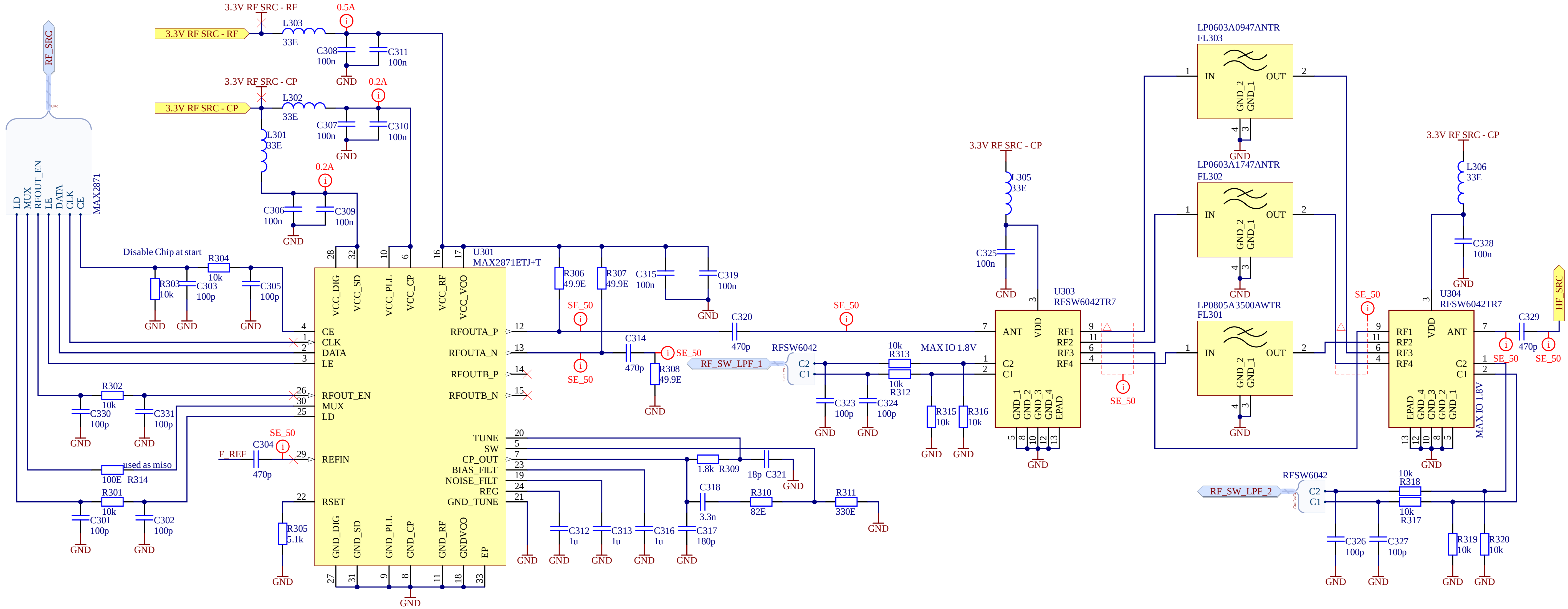
Design: ML

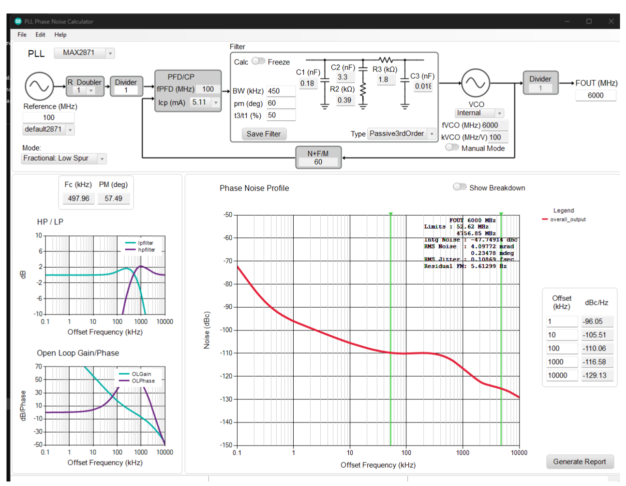
Check: RM

File: https://github.com/ml3rc/SNA.git

2 MCU.SchDoc

See LibreVNA Schematics for reference





Rset = 5.1kOhm as per datasheet

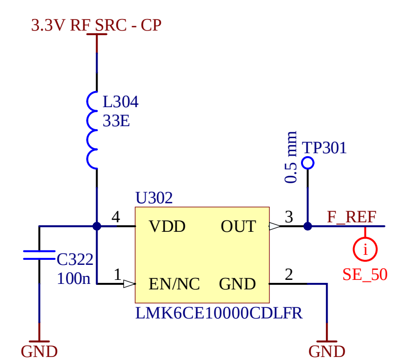
C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

The loop filter can be calculated with:
https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

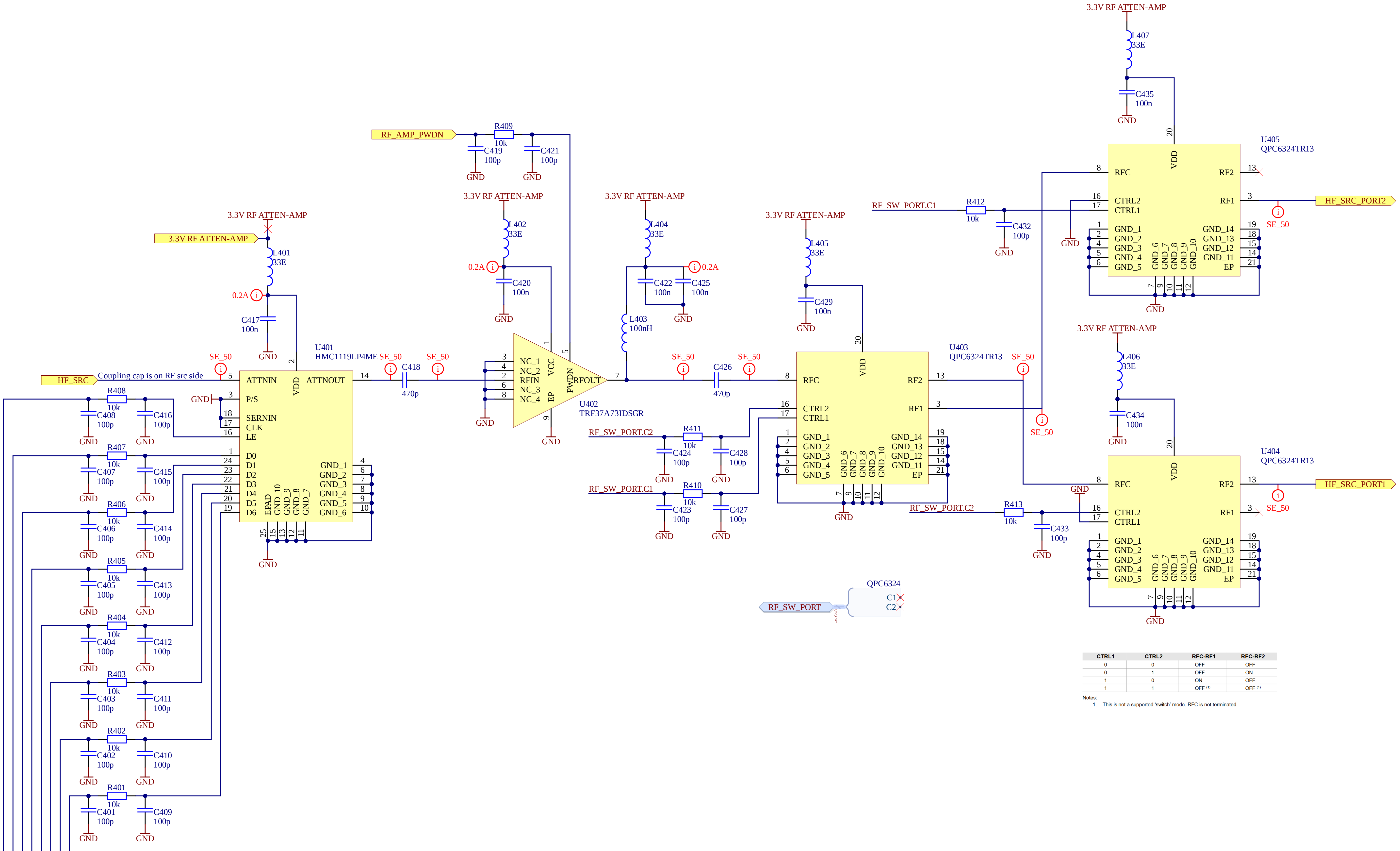
R_Crf_coupling = ((1)/(2*π*50MHz*470pF)) = 6.8Ohm

100MHz REF Clk.



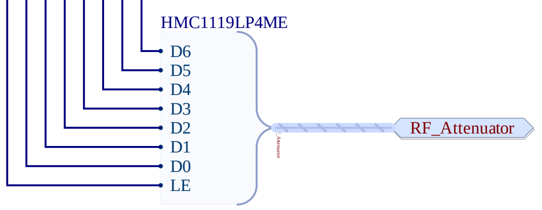
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	Project: SNA Board: SNA Page: RF Source	Number: SNA Variant: EMS Release: 03.09.2026	Git: 77bd9c41 Version: 1.0 Design: ML Check: RM
		Sheet 4 of 7 Size: A3	

File: https://github.com/ml3rc/SNA.git 3 RF Source.SchDoc



CTRL1	CTRL2	RFC-RF1	RFC-RF2
0	0	OFF	OFF
0	1	OFF	ON
1	0	ON	OFF
1	1	OFF [1]	OFF [1]

Notes:
1. This is not a supported 'switch' mode. RFC is not terminated.



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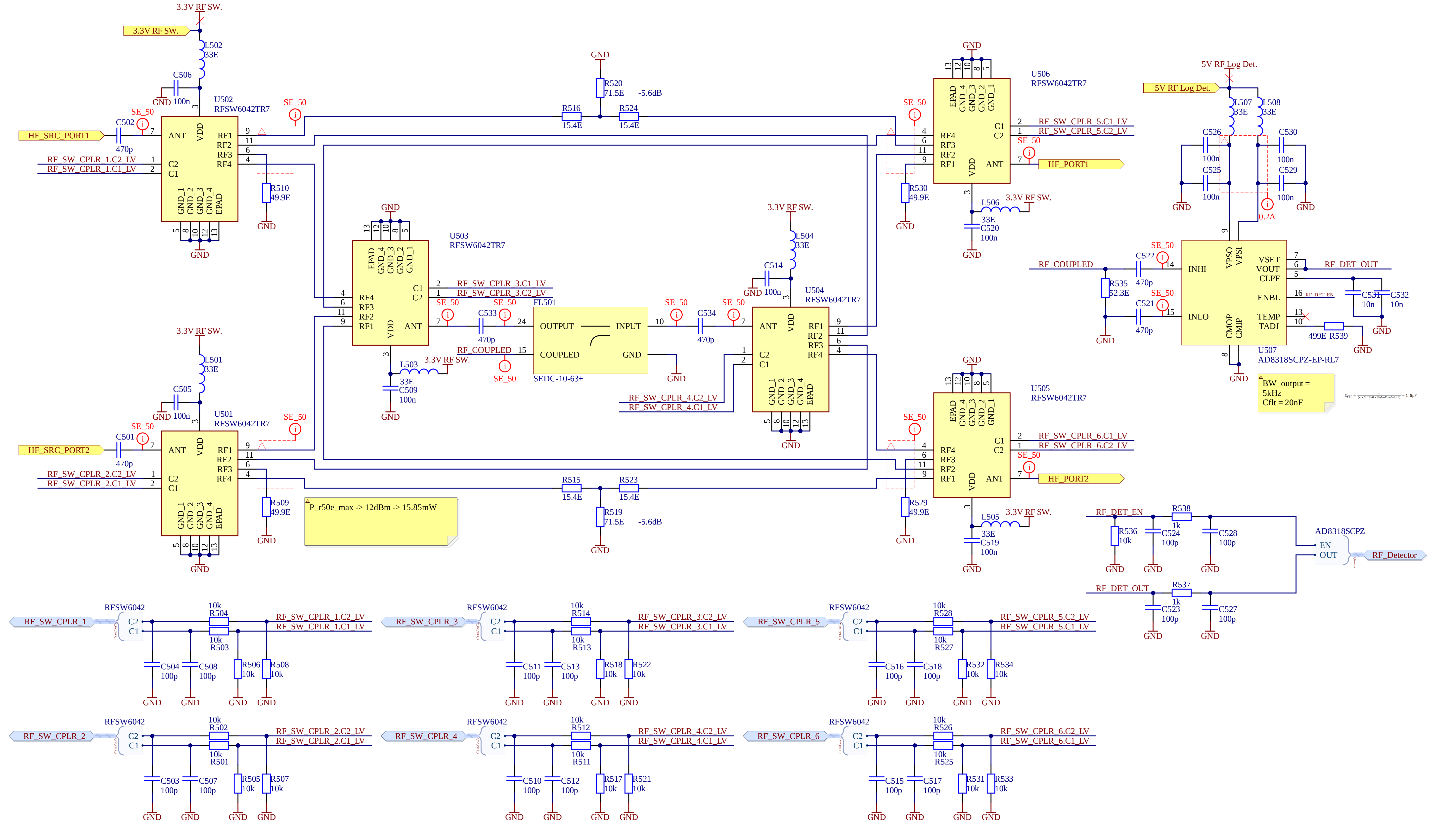
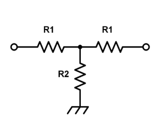
Gibb, Bern

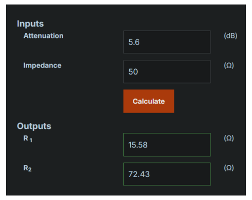
Project:	SNA	Number:	SNA
Board:	SNA	Variant:	EMS
Page:	RF Attenuation Amp	Release:	03.09.2026
Sheet	5 of 7	Size:	A3
Git:	77bd9c41	Version:	1.0
Design:	ML	Check:	RM

File: https://github.com/ml3rc/SNA.git   4 RF Attenuation Amp.SchDoc



T-Pad calculated with: https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/





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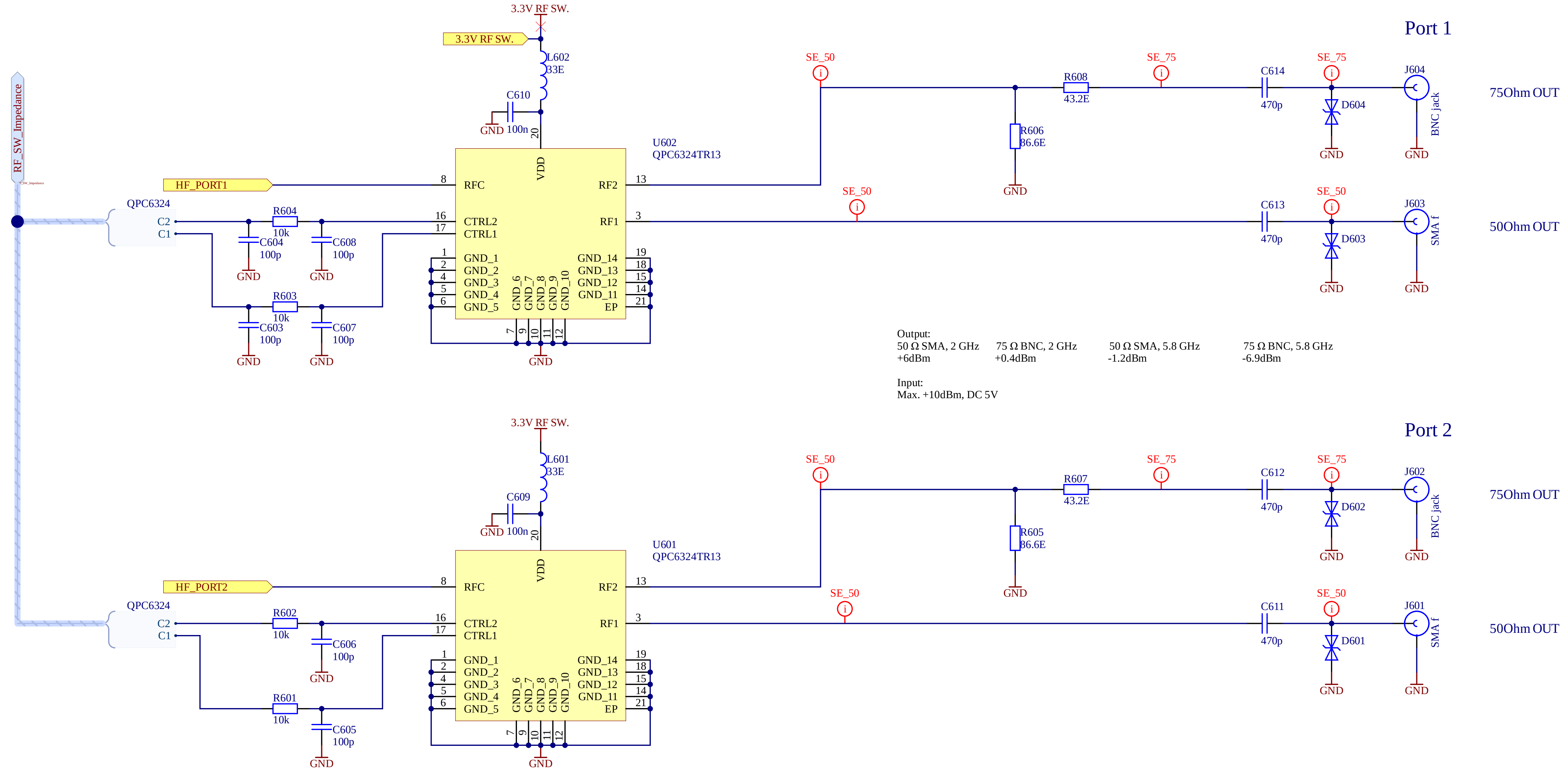
Gibb, Bern

Project: SNA	**Number:** SNA		**Git:** 77bd9c41
Board: SNA	**Variant:** EMS		**Version:** 1.0
Page: RF Directional Coupler	**Release:** 03.09.2026	**Sheet** 6 **of** 7	**Design:** ML
		Size: A3	**Check:** RM

File: https://github.com/ml3rc/SNA.git 5 RF Directional Coupler.SchDoc



50-75 Matching Pad:
solve({75=((1)/(((1)/(Rp))+((1)/(50))))+Rs,50=((1)/(((1)/(75+Rs))+((1)/(Rp))))})
Rp = 86.6Ohm
Rs = 43.3Ohm
Insertion Loss = 5.72dB

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Gibb, Bern

Project:	SNA	Number:	SNA
Board:	SNA	Variant:	EMS
Page:	RF Ports	Release:	03.09.2026
Sheet	7 of 7	Size:	A3
Git:	77bd9c41	Version:	1.0
Design:	ML	Check:	RM

File: https://github.com/ml3rc/SNA.git 6 RF Ports.SchDoc